Organization-Aware, Demand-Driven Shared AI Intelligence

A Research Proposal for Cost-Efficient, Secure, and Adaptive Knowledge Reuse Across Organizational Applications

Research Concept Paper

Software Engineering • Artificial Intelligence Systems • Cloud Computing • Systems Optimization

Preliminary proposal — not yet experimentally validated

Central idea
A shared intelligence layer that selectively creates, reuses, updates, and serves knowledge artifacts across an organization's authorized applications and AI agents.

Core question
Can demand-driven reuse reduce total cost and latency while preserving answer quality, information freshness, and access-control guarantees?

Prepared for preliminary academic discussion. The proposed research gap and contributions require verification through a focused literature review.

Abstract

Organizations increasingly integrate generative AI into educational platforms, retail and point-of-sale (POS) systems, customer-support chatbots, internal business applications, and automated workflows. These applications often depend on overlapping information, including documents, policies, operational procedures, product knowledge, and recurring answers. When applications independently retrieve, prepare, and process this information, they may incur redundant computation, increased latency, repeated context preparation, and avoidable operating costs.

This proposal investigates Organization-Aware, Demand-Driven Shared AI Intelligence: a shared infrastructure layer that enables authorized applications, users, and AI agents to reuse appropriate knowledge artifacts and computational results across an organization. Rather than preparing all potentially useful information in advance, observed user requests influence what should be processed, which artifacts should be retained, and when additional preparation is economically justified.

The proposed system considers multiple artifact types, including embeddings, extracted evidence, summaries, validated answers, and task-specific representations. Decisions account for expected demand, processing cost, reuse potential, freshness, answer quality, access permissions, and maintenance overhead. A university knowledge assistant and a customer-support chatbot are proposed as initial evaluation scenarios. The study will compare upfront preparation, independent on-demand processing, fixed reuse policies, and adaptive demand-driven reuse.

Expected contribution: a cost-aware and security-conscious framework for evaluating shared knowledge reuse across applications, supported by an economic model, an adaptive decision policy, and empirical measurements. The novelty of this combination remains to be established through a systematic review of existing research.

Keywords: Shared AI Intelligence, Retrieval-Augmented Generation, Demand-Driven Processing, Knowledge Reuse, Multi-Tenant Systems, AI Agents, Semantic Caching, Cloud Cost Optimization, Access Control.

1. Introduction

1.1 Background

Generative AI applications increasingly rely on external information to answer domain-specific questions, summarize material, support decisions, and automate workflows. A university assistant may retrieve lecture notes and academic policies; a support chatbot may consult product documentation and approved answers; a POS application may use product manuals and store procedures.

Although these applications serve different purposes, they may repeatedly access the same sources or perform similar information-processing tasks. If each application implements its own ingestion, embedding, retrieval, summarization, caching, and model interaction, useful work may be duplicated.

This motivates treating knowledge preparation and reuse as shared organizational infrastructure rather than as an isolated responsibility of each application. However, centralizing information alone does not guarantee lower cost: storage, coordination, security enforcement, maintenance, and unused preparation can offset the benefits.

1.2 Motivation

Upfront preparation can reduce future request latency, but may waste resources on rarely accessed information. Independent on-demand processing avoids some unnecessary preparation, but repeated requests may pay for overlapping work. A demand-driven shared approach offers a possible middle ground: requests can trigger processing when needed, and useful results can be retained for subsequent authorized requests.

The system must still account for uncertain future demand, first-request latency, source updates, and the possibility that derived information becomes outdated or inappropriate to share.

1.3 Research Perspective

The project is approached primarily from software engineering, systems design, and business economics. It does not assume that a new foundation model is necessary. Instead, it investigates whether orchestration, artifact selection, resource allocation, and lifecycle management can improve existing AI capabilities.

2. Problem Statement

Organizations adopting AI-assisted applications face several connected challenges:

- Redundant computation: overlapping documents, evidence, summaries, or answers may be processed repeatedly.
- Uncertain demand: preparing everything can waste resources, while preparing only after requests arrive can increase latency and repeated work.
- Fragmented infrastructure: useful artifacts and optimizations may remain isolated inside individual applications.
- Knowledge lifecycle: source material changes, requiring refresh, invalidation, or recomputation.
- Heterogeneous workloads: different applications need different artifacts and response-time guarantees.
- Organizational security: shared infrastructure must preserve tenant boundaries, roles, provenance, and retention rules.
- Distributed costs and benefits: the request that triggers preparation may not be the only request that benefits.

The core problem is how to minimize the total cost and latency of AI-assisted information processing by selectively creating and reusing knowledge artifacts across applications while maintaining quality, freshness, and authorization guarantees.

3. Research Aim and Objectives

3.1 Aim

To investigate whether an adaptive, organization-aware shared intelligence layer can improve the economic and operational efficiency of AI-assisted applications through demand-driven knowledge preparation and controlled cross-application reuse.

3.2 Objectives

1. Review research on RAG, inference and semantic caching, reusable knowledge artifacts, multi-tenant architectures, and adaptive resource allocation.
2. Identify which information-processing tasks can be shared safely and economically across applications.
3. Develop an economic model comparing upfront preparation, independent on-demand processing, and shared demand-driven processing.
4. Design a policy for choosing between artifact creation, reuse, extension, refresh, recomputation, and direct querying of authoritative sources.
5. Define provenance and authorization requirements for sharing knowledge across users, applications, departments, and agents.
6. Evaluate the approach using contrasting organizational workloads.
7. Identify conditions under which shared intelligence is beneficial, neutral, or less effective than alternatives.

4. Proposed Concept and Architecture

The proposed architecture places a shared intelligence layer between organizational information sources and the applications that consume their knowledge. Applications retain their interfaces, business logic, and domain-specific operations; the shared layer provides reusable processing capabilities, artifact management, and adaptive reuse decisions.

Layer	Responsibilities
Organizational sources	Documents, manuals, policies, transcripts, FAQs, and authoritative databases/APIs.
Event-driven pipeline	Responds to source changes, user queries, identified knowledge gaps, and expiration events; coordinates asynchronous jobs.
Shared artifact store	Stores embeddings, evidence, summaries, validated answers, and other source-linked representations.
Adaptive decision policy	Chooses whether to reuse, create, extend, refresh, recompute, query directly, or decline/escalate.
Authorization and provenance	Enforces access rules and tracks source identity, versions, and applicable permissions.
Application/agent interface	Provides controlled access to shared knowledge for authorized apps and agents.

4.1 Event-Driven Processing

Events may include adding or modifying a source, receiving a query, identifying a missing artifact, observing repeated demand, changing permissions, or reaching an expiration condition. Events initiate evaluation; they do not automatically justify expensive processing. Queues may coordinate extraction, embedding, summarization, and validation jobs, with deduplication and idempotency to avoid duplicate work.

4.2 Artifact Types

Artifact	Potential benefit	Important limitation
Embeddings	Efficient semantic retrieval	Do not establish correctness by themselves
Extracted evidence	Reuse of relevant source passages	Must preserve provenance and permissions
Summaries	Reduce repeated context preparation	May omit detail or become outdated
Validated answers	Fast handling of stable recurring questions	Need grounding, versioning, invalidation
Retrieval results	Reduce repeated search work	Depend on query and source freshness
Workflow/agent plans	Reduce repeated planning for known tasks	Must validate current inputs and permissions

The study will not assume that all artifact types should be implemented simultaneously. A key question is which artifacts provide enough benefit to justify their creation and maintenance costs.

4.3 Authorization and Provenance

Every artifact should retain source identity, version, and relevant authorization metadata. Access must be checked when an artifact is used, not inferred from the permissions of the person who triggered its creation. Derived artifacts can combine information from several sources, so their authorization scope must reflect the information they expose. Source updates and permission revocations should trigger appropriate invalidation or recomputation.

4.4 Application and Agent Interface

Applications may access search, evidence-backed answers, artifact lookup, knowledge-gap detection, preparation requests, and provenance/freshness checks. The shared layer supplies reusable intelligence, not unrestricted authority over business systems. Applications remain responsible for domain-specific correctness.

5. Demand-Driven Knowledge Construction

5.1 From Document-Centric to Demand-Aware Processing

A newly uploaded document might initially receive minimal indexing. If repeated requests reveal that a particular section is important, the system may prepare additional evidence, summaries, or retrieval structures. This can avoid preparing every possible artifact before demand is known.

Demand does not automatically justify permanent storage. The system must estimate usefulness, expected reuse, reliability, permission scope, and ongoing cost.

5.2 Incremental Knowledge Growth

- Source facts: information supported by authoritative sources.
- Derived artifacts: summaries, evidence sets, and other generated representations.
- Observed demand: questions and patterns indicating what users need.
- User-provided claims: information that may require verification.
- Validated organizational knowledge: information approved for reuse under organizational rules.

Repeated questions are evidence of demand, not evidence that an answer is correct. Question patterns can reveal knowledge gaps, but source verification or human approval may be required before new information becomes trusted shared knowledge.

5.3 Cross-Application Reuse

An artifact prepared for one application may benefit another if both are authorized to use the underlying information and the artifact meets their requirements. For example, a verified product explanation may support both a support chatbot and a staff assistant. A derived artifact containing restricted customer information must not become globally available merely because its format is reusable.

6. Potential Application Domains

6.1 Education and University Systems

A university assistant may support students, instructors, and administrators using lecture materials, academic policies, course documents, and approved institutional information. Repeated questions can indicate where additional preparation is useful. Key constraints include course-specific access, restricted assessments, and changing academic policies.

6.2 POS and Retail

Retail systems may reuse product documentation, troubleshooting instructions, staff procedures, and store policies. However, live inventory, prices, transaction state, and customer balances must come from authoritative operational systems. Shared AI knowledge must not replace transactional consistency.

6.3 Customer-Support Chatbots

Recurring questions about product features, policies, and troubleshooting may justify validated reusable answers or evidence bundles. Novel questions may require additional retrieval or reasoning. Popularity alone must never determine correctness, and answers must be refreshed when source policies change.

6.4 Internal Applications and AI Agents

Multiple internal tools and agents may repeatedly consult the same documentation or construct similar task plans. Shared artifacts may reduce repeated work, but agents must validate current inputs and permissions before using a reusable plan or taking action.

6.5 Cross-Domain Generalization

The domains differ in request overlap, information stability, security requirements, and latency constraints. Experiments should test whether a shared decision policy generalizes or requires domain-specific configuration.

7. Research Questions

1. RQ1 — Cross-application reuse: Under what conditions can artifacts prepared for one application reduce the cost of another authorized application?
2. RQ2 — Demand-driven preparation: Can observed request patterns guide artifact creation more efficiently than upfront preparation or independent on-demand processing?
3. RQ3 — Artifact selection: Which representations provide the best trade-off among creation cost, reuse benefit, storage, freshness, and answer quality?
4. RQ4 — Adaptive policy: Can adaptive decisions outperform fixed reuse rules under changing workloads?
5. RQ5 — Economic allocation: How should preparation costs and shared benefits be evaluated when one request triggers work that benefits many users?
6. RQ6 — Knowledge lifecycle: How should source changes, conflicting information, and permission revocation affect artifacts?
7. RQ7 — Organizational security: Can cross-application reuse preserve authorization boundaries and prevent information leakage?
8. RQ8 — Generalization: Do benefits observed in educational and support workloads extend to other domains?

The first empirical study should prioritize a small subset of these questions rather than attempt to answer all of them simultaneously.

8. Economic Model

8.1 Basic Comparison

Let N be the number of requests that can benefit from an artifact; C_p the preparation cost; C_q the per-request cost without the artifact; C_r the per-request cost when reused; and C_m the maintenance cost over the evaluation period.

$$C_{independent} = N \cdot C_q$$

$$C_{shared} = C_p + N \cdot C_r + C_m$$

Shared preparation is beneficial under this simplified model when:

$$N(C_q - C_r) > C_p + C_m$$

If $C_q > C_r$, the approximate break-even request count is:

$$N^* = (C_p + C_m) / (C_q - C_r)$$

This model is a starting point. A realistic evaluation must include storage, orchestration, retrieval overhead, failed processing, concurrency, access-control checks, refresh frequency, model prices, and answer quality. Monetary expenditure and latency should be measured separately because a strategy may improve one while worsening the other.

8.2 Shared Costs and Benefits

The request that triggers preparation may not be the only request that benefits. The study should distinguish whether sharing reduces total expenditure, how shared costs might be allocated among users or applications, and whether creating the artifact is justified at all. An allocation policy can be unfair even when the overall architecture is efficient.

8.3 Optimization Constraints

Cost reduction cannot be pursued at the expense of critical requirements. Outdated answers, unauthorized disclosure, or degraded correctness are not acceptable trade-offs. The decision policy must operate within explicit quality, freshness, and authorization constraints.

9. Related Work and Preliminary Research Gap

9.1 Retrieval-Augmented Generation

RAG retrieves external information and incorporates it into a model's context. It provides a foundation for organizational assistants, but repeated retrieval and context preparation may still incur overhead.

9.2 Inference and Knowledge Caching

RAGCache investigates caching techniques for retrieval-augmented generation. CacheBlend studies reuse of model-side computation across retrieved text chunks. These works establish that reuse can extend beyond storing final answers.

9.3 Semantic Caching and Multi-Tenant Infrastructure

Semantic caching investigates reuse of results for sufficiently similar requests, while multi-tenant AI architectures examine shared infrastructure under authorization and operational constraints. Both areas are relevant, but similarity alone does not guarantee that an answer remains correct or applicable.

9.4 Preliminary Gap

The proposal does not claim that shared knowledge, demand-driven processing, semantic caching, or multi-tenant AI are individually new. The potential gap to investigate is whether a unified decision framework can manage multiple types of artifacts across authorized applications while jointly considering demand, preparation cost, source changes, answer quality, provenance, and authorization boundaries.

A systematic review must determine whether existing systems already address this combination. If comparable systems exist, the contribution should be narrowed to a specific limitation or a cross-domain evaluation.

10. Research Methodology

10.1 Research Design

The study will use a comparative systems-evaluation approach, beginning with a literature review and cost model, followed by workload design and either a simulator or a controlled prototype. Simulation may be sufficient for exploring demand distributions and break-even conditions; claims about actual latency, model expenditure, and answer quality require measurements from a real system.

10.2 Baseline Strategies

1. Upfront preparation: eligible artifacts are prepared before demand is observed.
2. Independent on-demand processing: each request performs required work without cross-request reuse.
3. Fixed reuse policy: artifacts are reused under predefined caching, expiration, or preparation rules.
4. Adaptive demand-driven reuse: the system chooses whether to prepare, reuse, extend, refresh, or recompute based on workload and cost estimates.

Baselines should be compared fairly, with equivalent answer-quality requirements and the same source data.

10.3 Evaluation Scenarios

- Organizational knowledge assistant: questions over documents and policies, with role-based access and changing sources.
- Customer-support chatbot: recurring and novel questions, policy-dependent answers, approved reusable responses, and escalation requirements.
- Optional POS workload: stable reference knowledge separated from live transactional queries.

10.4 Workload Design

- Number of users and applications; request frequency and overlap.
- Source size and artifact creation cost; source update rate.

- Frequency of unique questions and semantic similarity between requests.
- Authorization boundaries and response-time requirements.
- Model and infrastructure cost/performance characteristics.

Synthetic workloads provide control, but realistic traces or carefully designed representative workloads would strengthen external validity.

10.5 Evaluation Metrics

Dimension	Metrics
Economics	Total cost; cost per correctly answered query; preparation cost
Performance	End-to-end latency; time to first token; throughput
Reuse	Artifact reuse rate; avoided recomputation; cache hit rate
Answer quality	Correctness; evidence support; retrieval quality
Freshness	Stale-answer rate; update propagation time
Security	Unauthorized access; permission violations; revocation effectiveness
Resources and operations	Storage footprint; compute/network use where measurable; failed jobs; queue delays

10.6 Analysis and Reproducibility

The study should compare strategies under the same workloads and report sensitivity to request overlap, update frequency, preparation cost, and artifact lifetime. Multiple runs and uncertainty estimates should be used where feasible. Authorization tests should include source permission changes, revocation, and attempts to access derived artifacts outside the permitted scope.

The workload assumptions, artifact definitions, model and infrastructure configuration, cost calculations, and evaluation procedures should be documented so results can be reproduced.

11. Expected Contributions

1. Economic framework: compare independent processing, upfront preparation, fixed reuse, and demand-driven reuse.
2. Adaptive decision policy: select among creation, reuse, extension, refresh, and recomputation.
3. Shared intelligence architecture: expose reusable knowledge capabilities across authorized applications without centralizing all domain logic.
4. Comparative evaluation: identify workloads that benefit, and conditions where sharing introduces costs or risks.
5. Security and lifecycle guidance: address provenance, authorization, updates, invalidation, and promotion of interaction-derived information.
6. Cross-domain findings: assess whether the decision policy generalizes across organizational settings.

These are potential contributions, not completed results. Novelty and significance depend on the literature review and experimental evidence.

12. Scope and Limitations

The initial study will focus on document-derived knowledge, recurring information needs, and a limited set of reusable artifacts. It will not initially attempt to develop a new foundation model, build a universal autonomous-agent platform, or optimize every organizational workflow. More complex media processing, cross-region placement, model-side cache reuse, and advanced cost allocation can be considered later.

- Results may depend strongly on workload patterns and request overlap.
- Future demand is difficult to estimate accurately.
- Model and infrastructure costs change over time.

- Open-ended answer correctness is difficult to measure perfectly.
- Sharing may provide limited benefit when applications use different information or have incompatible permissions.
- Shared infrastructure introduces operational complexity that may offset savings.
- Simulated workloads may not fully represent real organizational behavior.

13. Proposed Work Plan

Phase 1 — Literature review	Verify the closest work and build a related-work comparison matrix.
Phase 2 — Research refinement	Narrow the research question and state evidence-supported novelty claims.
Phase 3 — Economic modeling	Define workload variables, costs, constraints, and break-even analysis.
Phase 4 — Experimental design	Select scenarios, data, baselines, workloads, and metrics.
Phase 5 — Prototype or simulation	Compare strategies and collect measurements.
Phase 6 — Analysis and reporting	Test hypotheses, report trade-offs, and document limitations.

14. Conclusion

Organizations increasingly deploy AI across applications that depend on overlapping information. Independent processing can create redundant computation and expense, while comprehensive upfront preparation can waste resources on knowledge that is rarely used. Shared knowledge infrastructure offers an opportunity to reuse useful work, but the benefits depend on demand, preparation cost, maintenance, security, and the nature of the information.

This proposal investigates an organization-aware, demand-driven shared intelligence layer that selectively creates, reuses, extends, refreshes, and invalidates knowledge artifacts across authorized applications and AI agents. Education, retail/POS, customer support, internal business applications, and agent workflows provide possible evaluation domains, but they should not be assumed to have identical requirements.

The central contribution, if validated, would be a cost-aware and security-conscious method for managing reusable AI knowledge under changing demand and organizational constraints. Existing work already covers important components of this vision; therefore, the project must establish its specific novelty through a rigorous literature review and demonstrate value through fair comparisons, measurable results, and transparent limitations.

The immediate priority is to identify a precise research gap and test whether an adaptive shared-intelligence approach solves it better than relevant existing methods.

Preliminary References

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3. Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems. https://arxiv.org/abs/2005.11401
4. Further literature to verify: semantic caching; reusable context compression; document-side inference reuse; multi-tenant information retrieval; cache invalidation; adaptive cloud resource allocation.

Academic integrity note: This document is a research proposal, not a report of completed experiments. No empirical performance improvement or definitive novelty claim is asserted. Bibliographic details and closely related work should be verified before submission.